

Equilibrium structure and spectroscopic constants of maleic anhydride

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Abstract The quadratic, cubic, and semi-diagonal quartic force fields of maleic anhydride have been calculated at the MP2 level of theory employing the cc-pVTZ basis set. The spectroscopic constants derived from the force field are in excellent agreement with the corresponding experimental values. The semi-experimental equilibrium structure has been derived from experimental ground state rotational constants and rovibrational corrections calculated from the cubic force field. This semi-experimental equilibrium structure is in excellent agreement with the ab initio structures computed at the CCSD(T) level of theory and it is closer to the ab initio structure than the purely experimental (or empirical) structures r_0 , $r_m^{(1)}$, and $r_m^{(2)}$ obtained by microwave spectroscopy as well as the equilibrium structure derived from gas-phase electron diffraction data.

Keywords Maleic anhydride · Semi-experimental equilibrium rotational constants · Semi-experimental equilibrium structure · Structure from CCSD(T) calculations · MP2/cc-pVTZ calculations of quadratic, cubic, and semi-diagonal quartic force fields · Rovibrational corrections to rotational constants · Anharmonic frequencies

Introduction

Maleic anhydride or dihydro-2,5-dioxofuran, $c\text{-C}_2\text{H}_2(\text{CO})_2\text{O}$, has many uses in industry because of its ready availability and its bifunctional reactivity. It is the simplest cyclic

anhydride with a five-membered ring. Its structure has been determined by single-crystal X-ray diffraction [1] and neutron diffraction in the solid state [2]. In the gas phase, an empirical substitution structure was derived from the ground state rotational constants of 10 isotopologues [3] and, quite recently, the molecular structure was reinvestigated by the gas-phase electron diffraction (GED) method and an equilibrium structure was obtained fixing the values of the C–H bond length and the $\angle(\text{C}–\text{C}–\text{H})$ bond angle as well as the difference between the C=C and C–O bond lengths at their ab initio MP2/aug-cc-pVTZ values [4].

The vibrational spectrum has been studied in gas, liquid, and solid phases using infrared, Raman, and inelastic neutron scattering spectroscopies [2, 5, 6]. The frequencies of the ring-bending and ring-twisting vibrations were also determined from relative intensity measurements of the vibrationally excited rotational transitions [7]. The ring-bending potential function was obtained from the variation of the rotational constants with the ring-bending quantum number [8]. From the form of the potential as well as from line intensity alternations of the rotational transitions it was possible to confirm that the molecule is planar with C_{2v} symmetry. In the same work, the electric dipole moment was measured. Finally, the molecular Zeeman effect was also observed permitting the determination of the diagonal elements of the molecular rotational g -tensor [9].

The geometrical structure of maleic anhydride is completely defined by eight parameters. With 10 isotopologues available, 30 ground state inertial moments are known. They are independent because the inertial defect relation for planar molecules, $\Delta_0 = I_c - I_b - I_a = 0$, cannot be strictly applied to ground state moments; if it could, it would reduce the number of independent observations from 30 to 20. The ground state moments are, however, composed not only of the large part due to the geometrical

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structure but also of a (much) smaller part due to the rovibrational (and possibly other) contributions which precludes the application of the inertial defect relation.

If this latter small part can be calculated by ab initio methods (as discussed in sections “[Computational details](#)”, “[Ab initio structure](#)”, “[Anharmonic force field](#)”) and subtracted from the ground state inertial moments, we obtain the semi-experimental inertial moments which by a least-squares fit (implicitly using the above inertial defect relation) yield the semi-experimental equilibrium structure (as discussed in “[Semi-experimental equilibrium structure](#)” section).

If, however, this latter part is simply modeled using a number of additional parameters, we obtain the empirical structures which are named after the models used (“[Empirical structures](#)” section). These additional “rovibrational” parameters must then be determined together with the structural parameters by a least-squares fit to the ground state inertial moments.

The actual purpose of the above ab initio method is to compute a force field up to cubic terms from which the rovibrational contributions can be derived, a task which can now be performed routinely. Furthermore, useful quantities are known, the values of the quartic centrifugal distortion constants and also values for the variation of the inertial moments with the vibrational quantum numbers. Therefore, maleic anhydride seems to be a good example of a molecule for which it should be possible to determine an accurate semi-experimental equilibrium structure and also to assess the quality of the empirical structures.

Computational details

The ab initio structure was optimized with the coupled cluster method with single and double excitations [10] and a perturbative treatment of connected triples [11] [CCSD(T)] with the correlation-consistent polarized covalence triple zeta (cc-pVTZ) basis set [12]. This calculation was performed with the MOLPRO program package [13]. The coupled cluster T_1 diagnostic [14] was also calculated at the CCSD(T)/cc-pVTZ level of theory. To check the convergence, some calculations were also performed with the second-order Møller–Plesset perturbation theory (MP2) [15] using the cc-pVTZ and cc-pVQZ basis sets [12]. To account for the electronegative character of the O atoms, the aug-cc-pVQZ basis set [16] was also used at the MP2 level. To estimate the contribution of the inner-shell correlation effects on the molecular geometry the correlation-consistent polarized weighted core-valence quadruple zeta (cc-pwCVQZ) [17] basis set was employed at the MP2 level with and without the frozen core approximation. It is indeed well established that the small

changes at the CCSD(T) level can be approximated rather well at the MP2 level [18, 19], at least for first-row atoms.

The anharmonic force field was calculated at the MP2 level of theory with the cc-pVTZ basis set. The optimized molecular geometry was calculated first. Then, the associated harmonic force field was evaluated analytically in Cartesian coordinates at the optimized geometry. The cubic and semi-diagonal quartic normal coordinate force constants were determined with the use of a finite displacement procedure involving displacements along reduced normal coordinates and the calculation of analytic second derivatives at these displaced geometries. Evaluation of the anharmonic spectroscopic constants was based on second-order rovibrational perturbation theory [20]. All the MP2 calculations were performed with the GAUSSIAN 03 (g03) program [21].

Ab initio structure

From the value of the T_1 diagnostic, 0.017 at the CCSD(T)/cc-pVTZ level, it may be concluded that the non-dynamical electron correlation is not overly important. Therefore, the CCSD(T) structure is expected to be reliable. The results of the ab initio geometry optimizations are given in Table 1. The effect of diffuse functions is quite small at the quadruple zeta level (the largest variation is an increase of the C=O bond by 0.0009 Å). What is interesting is that it slightly increases the CO bond lengths whereas the basis set enlargement cc-pVQZ → cc-pV5Z is expected to decrease them by about the same amount. In conclusion, at the quadruple zeta level without diffuse functions, the structure is accidentally close to the equilibrium structure. Correlating all the electrons at the MP2/cc-pwCVQZ gives the expected shortening of bonds (in Å): 0.0028 for C–O, 0.0022 for C=O, 0.0034 for C–C, 0.0033 for C=C, and 0.0015 for C–H. As usual, the effect on the angles is quite small (smaller than 0.03°) and can thus be neglected. Finally, the best ab initio structure is calculated using the following approximation:

$$r_e^{\text{BO}} = \text{CCSD(T)/cc-pVTZ(fc)} + \text{MP2/cc-pVQZ(fc)} \\ - \text{MP2/cc-pVTZ(fc)} + \text{MP2/cc-pwCVQZ(ae)} \\ - \text{MP2/cc-pwCVQZ(fc)}$$

where fc means frozen core and ae all electrons correlated. For first-row atoms, this approximation was found to have an accuracy not much worse than 0.001 Å [16, 22, 23].

Anharmonic force field

The computed and experimental ground state quartic centrifugal distortion constants of the parent isotopologue are

Table 1 Structure of maleic anhydride (distances in Å and angles in degrees)

Method ^a	CCSD (T)	MP2(fc)			MP2 (fc)	MP2 (ae)	$r_e^{\text{BO c}}$	$r_e^{\text{se d}}$	ED + MW ^e
Basis set ^b	VTZ	VTZ	VQZ	AVQZ	wCVQZ	wCVQZ			
C–C	1.4922	1.4852	1.4833	1.4833	1.4828	1.4794	1.4869	1.4849(5)	1.485(1)
C=C	1.3382	1.3358	1.3334	1.3340	1.3330	1.3297	1.3325	1.3324(5)	1.332(1)
O–C	1.3901	1.3902	1.3877	1.3884	1.3874	1.3846	1.3848	1.3848(3)	1.386(1)
C=O	1.1947	1.1965	1.1938	1.1947	1.1934	1.1912	1.1898	1.1894(2)	1.192(1)
C–H	1.0786	1.0770	1.0764	1.0767	1.0763	1.0748	1.0765	1.0765(2)	1.078 ^f
C–C=C	107.932	108.0034	108.014	108.017	108.018	108.035	107.94	107.96(1)	107.8(1)
O=C–C	129.560	129.5709	129.626	129.657	129.629	129.621	129.62	129.67(3)	129.2(2)
C=C–H	129.867	129.6471	129.670	129.691	129.662	129.642	129.89	129.90(1)	129.7 ^f
O–C–C	107.798	107.8397	107.794	107.809	107.792	107.771	107.75	107.78(2)	108.2(1)
C–O–C	108.540	108.3137	108.384	108.348	108.382	108.388	108.61	108.52(3)	107.9(1)
O=C–O	122.641	122.5894	122.580	122.534	122.579	122.608	122.63	122.55(4)	122.6(1)

^a *fc* = frozen core, *ae* = all electron correlated^b VTZ = cc-pVTZ; VQZ = cc-pVQZ; AVQZ = aug-cc-pVQZ; wCVQZ = cc-pwCVQZ^c Born–Oppenheimer structure: $r_e^{\text{BO}} = \text{CCSD(T)/cc-pVTZ(fc)} + \text{MP2/cc-pVQZ(fc)} - \text{MP2/cc-pVTZ(fc)} + \text{MP2/cc-pwCVQZ(ae)} - \text{MP2/cc-pwCVQZ(fc)}$ ^d Semi-experimental structure, see text^e Combined analysis of GED data and semi-experimental rotational constants B_e , Ref. [4]^f Assumed at the MP2/aug-cc-pVTZ value

reported in Table 2. The experimental constants are reproduced with an accuracy of a few percent. Taking into account the fact that the experimental constants are ground state values whereas the computed constants are equilibrium values, the agreement is extremely good, indicating that the harmonic force field is likely to be accurate.

Table 3 contains the experimental and computed vibrational fundamental frequencies as well as the computed anharmonicity corrections. Most experimental values have been measured in the solid state only. Nevertheless, the agreement with the computed values is good, the median absolute deviation being only 8.9 cm^{-1} . This is again an indication that the anharmonic force field is reliable.

However, the most critical test is to check whether the rotation–vibration interaction constants are well reproduced

for the ring-bending and the ring-twisting modes. These two vibrations are indeed large amplitude vibrations with a non-negligible quartic contribution and it is therefore not obvious that the method used to calculate the cubic force field is accurate enough in such a case. The results are given in Table 4. The extremely good agreement gives us confidence that the cubic force field is accurate enough to calculate a semi-experimental equilibrium structure.

Semi-experimental equilibrium structure

The semi-experimental equilibrium rotational constants, B_e , can be calculated from the experimental ground state rotational constants, B_0 , using the following equation:

$$B_e = B_0 + \Delta B_{\text{vib}} + \Delta B_{\text{el}} \quad (1)$$

where ΔB_{vib} is the rovibrational correction calculated from the cubic force field and ΔB_{el} is the electronic correction which may be obtained from the rotational g tensor via the following formula:

$$\Delta B_{\text{el}} = -\frac{m_e}{M_p} g B_e \quad (2)$$

where m_e and M_p are the masses of the electron and the proton, respectively. The experimental values of g are known from Ref. [9].

Table 2 Experimental and computed quartic centrifugal distortion constants for maleic anhydride (in kHz)

	Calc ^a	Exp ^b	Exp – calc	Exp – calc (%)
Δ_J	0.113	0.116(1)	0.003	2.89
Δ_{JK}	0.227	0.241(5)	0.014	5.63
Δ_K	2.397	2.442(2)	0.045	1.84
δ_J	0.03167	0.0324(1)	0.001	2.25
δ_K	0.364	0.367(7)	0.003	0.72

^a From the MP2/cc-pVTZ harmonic force field^b Ground state values from Ref. [7]

Table 3 Computed vibrational frequencies, anharmonicity corrections ($\omega - \nu$), and experimental fundamental vibrational wavenumbers (cm^{-1}) for maleic anhydride

	$\nu(\text{calc})^{\text{a}}$	$\omega - \nu^{\text{a}}$	$\nu(\text{exp})^{\text{b}}$		$e - c^{\text{c}}$
			Gas	Solid	
A_1					
C–H sym stretch	3182.0	118.9		3132	−50
C=O sym stretch	1856.2	30.4	1865	1857	1
C=C stretch	1631.1	3.5		1595	−36
C–O sym stretch	1255.3	27.7	1232	1243	−12
C–H ip bend	1059.6	18.1		1066	6
C–C sym stretch	863.8	17.7		872	8
Ring def	623.9	8.5		641	17
C=O ip bend	400.8	3.8		415	14
A_2					
C–H oop bend	966.9	15.9		959	−8
Ring def	772.3	8.7		768	−4
C=O ip bend	263.3	1.8		275	12
B_1					
C–H asym stretch	3154.0	125.1	3120	3124	−30
C=O sym stretch	1795.4	35.5	1805	1785	−10
C–H ip bend	1308.6	30.9		1309	0
C–O asym stretch	1043.9	29.9	1055	1059	15
C–C asym stretch	887.1	28.1		896	9
Ring def	693.8	12.2	699	697	3
C=O ip bend	556.1	6.6	556	563	7
B_2					
C–H oop bend	846.7	13.6	840	839	−8
Ring def	633.3	10.0		631	−2
C=O oop bend	160.7	2.8	161	200	39
MAD ^d					8.9
σ^{e}					20

^a From the MP2/cc-pVTZ anharmonic force field^b From Ref. [2]^c Exp(solid) – calc^d Median absolute deviation^e Standard deviation

The electronic correction is often negligible but, in the particular case of maleic anhydride, it is important to take it into account. Indeed, the ground state inertial defect, $\Delta_0 = I_c - I_b - I_a = -0.0073 \text{ u}\text{\AA}^2$, is quite small but, when only the vibrational correction is taken into account, the semi-experimental equilibrium inertial defect is not much smaller, $\Delta_e = -0.0072 \text{ u}\text{\AA}^2$. On the other hand, when the electronic correction is also taken into account, Δ_e decreases (in absolute value) to $0.00057 \text{ u}\text{\AA}^2$. The ground state rotational constants, the rovibrational corrections, the semi-experimental rotational constants as well as the residuals of the fit are given in Table 5.

Table 4 Experimental and computed rotation-vibration interaction constants (MHz) for maleic anhydride

	Axis	Calc ^a	Exp ^b
Ring bending (ν_{10})	α^A	32.33	32.20
	α^B	–2.86	–2.93
	α^C	–2.96	–3.01
Ring twisting (ν_{21})	α^A	10.66	10.85
	α^B	–0.60	–0.61
	α^C	–2.08	–2.09

^a From the MP2/cc-pVTZ cubic force field^b From Ref. [7]

The semi-experimental equilibrium structure was calculated from a weighted least-squares fit of the semi-experimental equilibrium moments of inertia. Different weighting schemes were used. The results presented were obtained with the iteratively reweighted least-squares (IRLS) method using the Huber weighting, that is to say a weight equal to [24]

$$w = 1 \quad \text{if } |r|/s \leq 1.345$$

$$w = 1.345/(|r|/s) \quad \text{if } |r|/s > 1.345 \quad (3)$$

where r is the residual of the previous iteration and s the corresponding standard deviation calculated from the median absolute deviation (MAD)

$$\text{MAD} = \text{median}|r_i - \text{median}(r_i)| \quad (4)$$

and

$$s = \text{MAD}/0.6745 \quad (5)$$

When the errors follow the normal law, this estimate of the standard deviation is identical to the value obtained with the usual definition, the difference being that it is much less sensitive to outliers. Huber estimation with $c = 1.345$ is 95% as efficient as ordinary least squares when applied to Gaussian errors, yet it gives very good protection against outliers. Note that the chosen value corresponds to $1.345s = 2\text{MAD}$. However, it is worth noting that the different weighting schemes give compatible results.

Comparison of the semi-experimental structure with the ab initio structure of section “**Ab initio structure**” is pleasing, the differences being hardly larger than twice the (rather small) standard deviations. Likewise, the agreement with the previous equilibrium structure derived from GED data is also quite satisfactory.

Empirical structures

Watson et al. [25] have suggested and successfully applied a mass-dependent r_m model (in two versions) for the

Table 5 Experimental ground-state rotational constants (B_0), MP2/cc-pVTZ rovibrational corrections (ΔB), semi-experimental rotational constants (B_e) and residuals of the fit (MHz) for isotopic species of maleic anhydride

Isotopic species	Ground state ^a			Rovibrational correction			Semi-experimental			Residuals of the fit		
	A_0	B_0	C_0	ΔA	ΔB	ΔC	A_e	B_e	C_e	$A_e - A_{\text{calc}}$	$B_e - B_{\text{calc}}$	$C_e - C_{\text{calc}}$
Main	6842.489	2467.614	1813.628	49.78	14.18	11.16	6892.69	2481.86	1824.80	0.004	0.013	0.004
d ₂	6106.159	2423.551	1734.998	41.51	13.89	10.46	6148.00	2437.51	1745.48	−0.038	0.002	−0.002
¹³ C ₂	6841.501	2452.607	1805.442	41.51	13.89	10.46	6891.43	2466.65	1816.48	−0.007	0.008	0.002
¹³ C ₃	6691.282	2462.240	1799.954	48.19	14.11	11.03	6739.87	2476.42	1811.00	0.064	−0.001	0.001
¹⁸ O ₁	6688.643	2467.808	1802.755	49.17	13.97	11.01	6738.22	2481.85	1813.78	0.045	−0.004	−0.003
¹⁸ O ₆	6790.337	2354.397	1748.276	49.12	13.41	10.65	6839.88	2367.87	1758.94	−0.068	−0.009	−0.013
¹³ C ₂ d ₂	6104.567	2409.078	1727.440	41.40	13.67	10.34	6146.31	2422.81	1737.79	0.126	−0.025	−0.004
¹³ C ₃ d ₂	5993.203	2418.373	1723.126	40.31	13.83	10.35	6033.84	2432.27	1733.49	−0.028	−0.002	−0.004
¹⁸ O ₁ d ₂	5971.996	2423.742	1724.104	41.02	13.71	10.34	6013.33	2437.52	1734.45	−0.040	0.013	0.003
¹⁸ O ₆ d ₂	6058.195	2314.121	1674.540	41.05	13.17	10.03	6099.58	2327.35	1684.58	0.051	0.003	0.006

^a Ref. [3]

rovibrational contribution where the additional terms depend on the atomic masses as $m^{1/2}$ as required. The mass-dependent coordinates r_m are obtained by a fit to the experimental ground state moments of the three functions for each isotopologue

$$I_0^\alpha = I_m^\alpha + c_\alpha (I_m^\alpha)^{1/2} + d_\alpha (m_1 \cdot m_2 \cdot \dots \cdot m_N / M)^{1/(2N-2)},$$

$$I_m^\alpha = I_{\text{rigid}}^\alpha(r_m) \quad (6)$$

where N is the number of atoms and c_α and d_α are constants for each axis $\alpha = a, b, c$. Using I_m^α only, i.e., taking no account at all of the rovibrational contribution and omitting the last two terms, yields the r_0 structure, while the r_m model, using the terms up to the c_α term, yields the $r_m^{(1)}$ structure and using all terms, the $r_m^{(2)}$ structure. For each of the three possible structures, we have obtained the structural parameters and, depending on the model, also the rovibrational parameters c_α and d_α by fitting the functions of Eq. 6 to the 3×10 ground state inertial moments of [3], starting with unity-weighted ordinary least-squares fits and then switching to iterative reweighted least-squares fits with Huber weighting. The results are presented in Table 6 where they are compared to the Born–Oppenheimer and semi-experimental equilibrium structural values of Table 1, r_e^{BO} and r_e^{se} , respectively.

The condition number which estimates how near-singular the design matrix of the least-squares problem is, increases significantly with the number of parameters chosen for fitting the functions of Eq. 6, from r_0 over $r_m^{(1)}$ to $r_m^{(2)}$. For r_0 and r_e^{se} with equal numbers of parameters, the condition numbers are practically equal. Any column vector of the design matrix contains the derivatives of the observed inertial moments with respect to a particular determinable parameter. Near-singularity results from the

fact that these column vectors differ progressively less in “direction”, i.e., are nearer to collinearity in the “space of observations”, when the number of column vectors increases with the number of parameters to be determined, while the dimension of the space of observations remains equal. This greatly impairs the reliability of the parameters obtained despite the decrease of the standard deviation, σ , from r_0 over $r_m^{(1)}$ to $r_m^{(2)}$. In fact, a lower σ will reliably indicate the better fit only when the number of determinable parameters in both fits is equal. Comparing the results for $r_m^{(1)}$ and $r_m^{(2)}$ exemplifies that the lower σ does not necessarily indicate the better fit because we know from experience the approximate magnitude of the rovibrational parameters c_α and d_α . While for $r_m^{(1)}$ the percentage of the rovibrational contribution to the inertial moments (see Eq. 6) is even somewhat below the most often encountered approximately 0.5%, it is roughly five times as large for $r_m^{(2)}$. Also, the opposite signs of c_α and the d_α point to a partial compensation of two very large components (the last and second to last terms of Eq. 6). The fit resulting in the semi-experimental $r_e^{\text{se}} = r_0(\text{se})$ structural values (“Semi-experimental equilibrium structure” section) is no doubt the by far best fit of all, judged by the standard deviation as well as the condition number. This is further confirmed by the lowest rms square deviations from the ab initio calculated structure as shown in Table 6. Of the empirical structures we prefer the $r_m^{(1)}$ structure. Belsley [26] has explained how the so-called variance-decomposition proportions (vdp’s; positive numbers <1) allow a more detailed assessment of the quality of the fits. When the highest condition index (the condition number) exceeds a certain numerical limit (e.g., 50), and the vdp’s of two (or more) of the parameters to be determined exceed 0.5 then these parameters are no longer sufficiently independent but

Table 6 Parameters of experimental and empirical structures, r_0 , $r_m^{(1)}$, $r_m^{(2)}$, contrasted to structures r_e^{BO} and r_e^{se} (distances in Å and angles in degrees) of maleic anhydride

Type of fit	r_e^{BO}	$r_e^{se} = r_0(se)$ IRLS	r_0 IRLS	$r_m^{(1)}$ IRLS	$r_m^{(2)}$ IRLS
Std. dev. fit		0.000837	0.00626	0.00279	0.000696
Cond. number		1091	1099	1308	3151
C–C	1.4869	1.4849(5)	1.487(3)	1.483(2)	1.4772(5)
C=C	1.3325	1.3324(5)	1.342(4)	1.335(3)	1.3453(8)
O–C	1.3848	1.3848(3)	1.390(2)	1.387(1)	1.3870(3)
C=O	1.1898	1.1894(2)	1.195(2)	1.195(1)	1.1886(3)
C–H	1.0765	1.0765(2)	1.074(1)	1.077(1)	1.0996(8)
C–C=C	107.94	107.96(1)	107.8(1)	107.9(1)	107.77(1)
O=C–C	129.62	129.67(3)	129.7(2)	129.6(1)	129.66(3)
H–C=C	129.89	129.90(1)	129.9(1)	130.0(1)	131.04(4)
O–C–C	107.75	107.78(2)	108.0(1)	108.1(1)	108.14(2)
C–O–C	108.61	108.52(3)	108.2(2)	108.1(1)	108.19(3)
O–C=O	122.63	122.55(4)	122.3(3)	122.3(1)	122.21(4)
rms diff.					
Bonds	0.0	0.0010	0.0050	0.0031	0.0125
Angles	0.0	0.056	0.247	0.300	0.556
c_a				0.038(10)	0.165(8)
c_b				0.046(13)	0.259(9)
c_c				0.058(14)	0.306(10)
d_a					−0.31(2)
d_b					−1.08(4)
d_c					−1.39(5)

near to linear dependency. Even when the numerical limits cited could rarely be applied to the determination of molecular structure by least squares fits to inertial moments (due to the inherent near-collinearity of the columns of the design matrix), we have observed for the $r_m^{(1)}$ structure only two vdp's over 0.6; for $r_m^{(2)}$, however, four vdp's over 0.9 and for the r_0 structure (and $r_0(se)$) three vdp's over 0.9.

In [25], Eq. 37, a method is suggested to correct the effect of the rotation of the principal axis system with respect to the contour of the molecule upon atomic substitution. The model changes from $r_m^{(1)}$, $r_m^{(2)}$ to $r_m^{(1r)}$, $r_m^{(2r)}$, respectively. Since the maximum rotation angle is no larger than 1.1 deg within the set of 10 isotopologues of this molecule, there is no need for this correction.

We have also checked if a correction for the Laurie contraction of the C–H bonds upon deuteration after [25], Eq. 31, is meaningful, changing the model from $r_m^{(1)}$, $r_m^{(2)}$ to $r_m^{(1L)}$, $r_m^{(2L)}$. But we have found either values smaller than their errors ($r_m^{(1L)}$) or fully unreasonable values ($r_m^{(2L)}$) with high correlation of bond length and amount of contraction.

In conclusion, we consider the r_e^{se} structure to be by far the best of all least-squares spectroscopic structures discussed so far for this molecule and the $r_m^{(1)}$ structure the best of all empirical structures with the sole advantage over r_e^{se} that it can be directly obtained from the ground state inertial moments without any further processing (as in “Computational details” section up to “Semi-experimental

equilibrium structure” section). The principal advantages of the spectroscopic equilibrium structure r_e^{se} in comparison to that derived from GED data is that it does not require any geometrical assumptions and has higher accuracy.

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